Elasticity

What is elasticity?

- *Elasticity* is a measure of the responsiveness to a change in a market condition.
- The concept applies to supply and demand.
- It measures the response to a change in:
  - The price of the good.
  - The price of a related good.
  - Income.

Price Elasticity of Demand

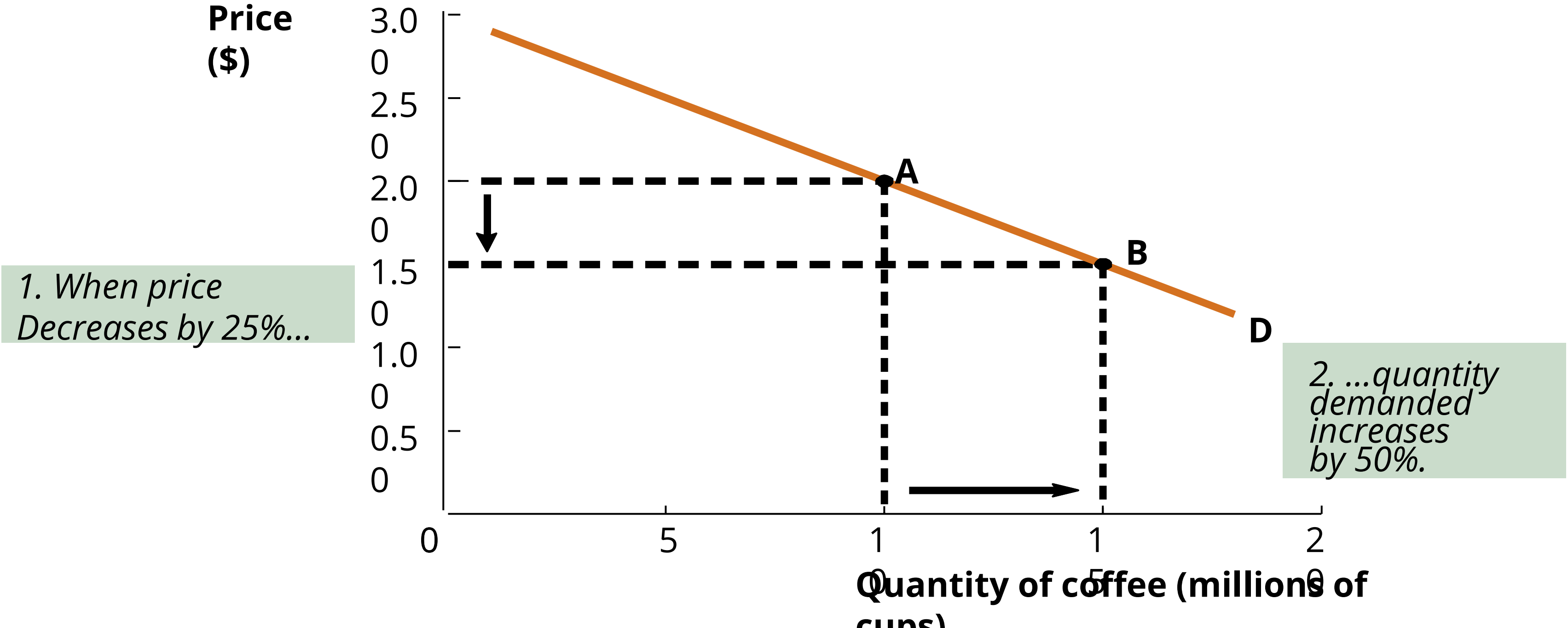
Price elasticity of demand is a measure of the extent to which the quantity demanded of a good changes when the price of the good changes.

We calculate the price elasticity of demand as follows:

$$\epsilon = \frac{\%\ change\ in\ quantity\ demanded\ of\ a\ good}{\%\ change\ in\ price\ of\ the\ good}$$

Price elasticity of demand

- The *price elasticity of demand* measures the magnitude of change in the quantity demanded from a change in its price. In other words, it estimates price sensitivity.



Calculating price elasticity

- The *midpoint method* calculates the elasticity at the midpoint of any two points. The formula is:

$$\varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{(Q_2 - Q_1)/[(Q_2 + Q_1)/2]}{(P_2 - P_1)/[(P_2 + P_1)/2]}$$

where point 1 is (P_1, Q_1) and point 2 is (P_2, Q_2).

- The midpoint elasticity is the difference of any two numbers divided by their average.

Calculating the price elasticity of demand

Find the price elasticity of demand using the midpoint formula for the points along a demand curve: ($10, 350) and ($20, 150).

Calculating the price elasticity of demand

Find the price elasticity of demand using the midpoint formula for the points along a demand curve: ($10,350) and ($20, 150).

$$\varepsilon_d = \frac{\%\Delta Q}{\%\Delta P} = \frac{(150-350)/[(150+350)/2]}{(\$20-\$10)/[(\$20+\$10)/2]} = \frac{-200/250}{\$10/\$15} = -1.2$$

- Interchanging points yields the same answer.

$$\varepsilon_d = \frac{\%\Delta Q}{\%\Delta P} = \frac{(350-150)/[(350+150)/2]}{(\$10-\$20)/[(\$10+\$20)/2]} = \frac{200/250}{-\$10/\$15} = -1.2$$

- The elasticity is unitless.

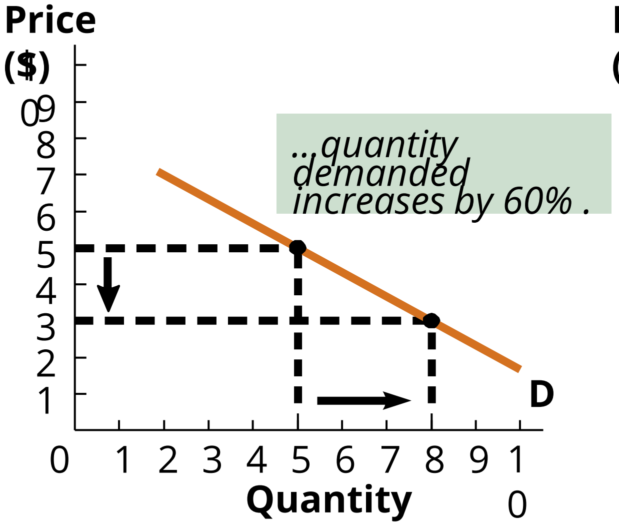
Categorizing elasticities

- All goods and services can be broadly categorized based on elasticity.
- The main categories are:
  - *Elastic*: A change in price causes a relatively *large* percentage change in quantity demanded.
  - *Inelastic*: A change in price causes a relatively *small* percentage change in quantity demanded.

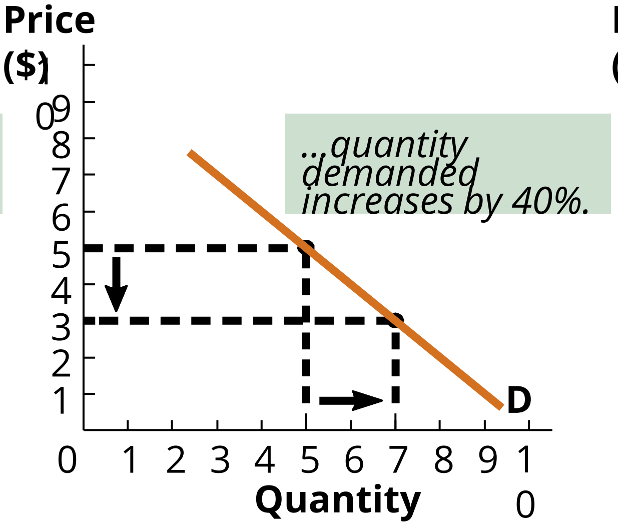
Categorizing elasticities

Between these two extremes, the elasticities are divided into three categories.

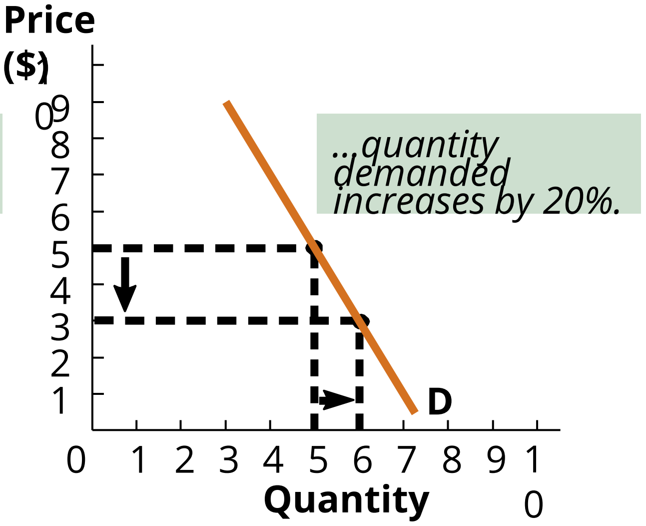
Elastic ($|\varepsilon_d| > 1$)



Unit-elastic ($|\varepsilon_d| = 1$)



Inelastic ($|\varepsilon_d| < 1$)

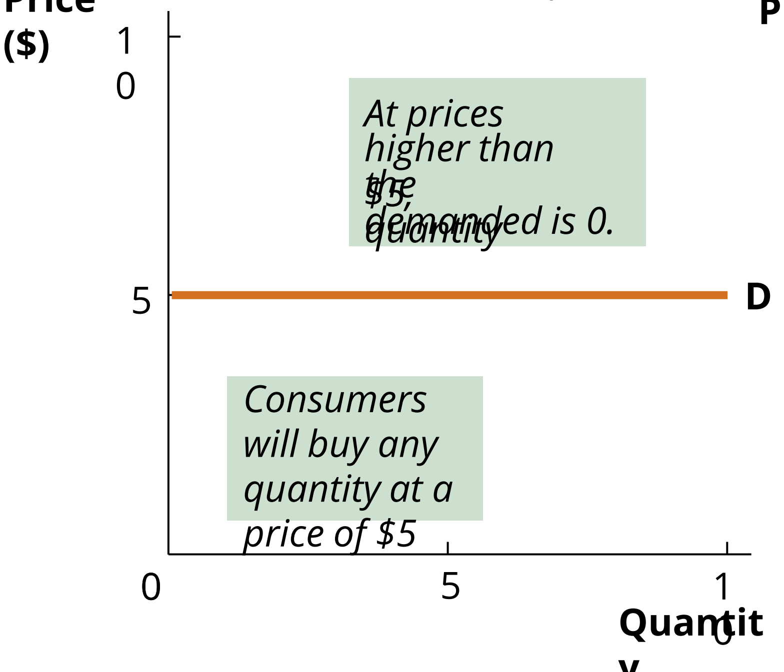


When price declines by the same amount (say 40%)…

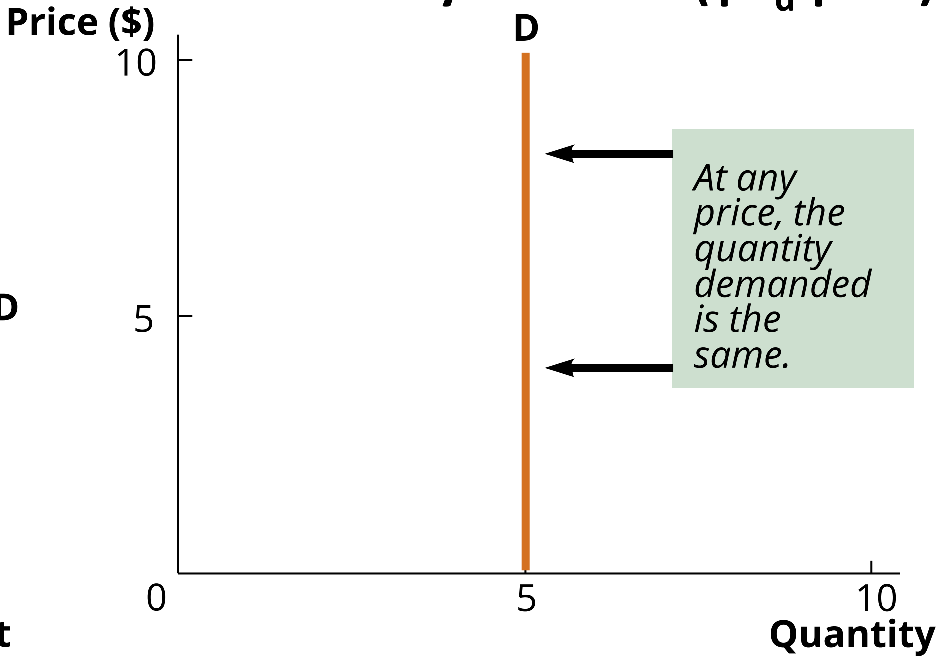
Categorizing elasticities

At the extremes, demand can be either *perfectly elastic* or *perfectly inelastic*.

Perfectly elastic ($|\varepsilon_d| = \infty$)



Perfectly inelastic ($|\varepsilon_d| = 0$)



Using price elasticity of demand

- Knowing whether the demand for a good is elastic or inelastic is extremely useful in business.
  - Allows managers to determine whether a price increase will cause *total revenue* to rise or fall.
- Total revenue is the amount that a firm receives from the sale of goods and services.
  - Total revenue (TR) equals price paid (P) multiplied by quantity sold (Q), or TR = P x Q.

Total Revenue and Elasticity

- If demand is elastic, a 1 percent price cut increases the quantity sold by more than 1 percent and *TR increases.*
- If demand is inelastic, a 1 percent price cut increases the quantity sold by less than 1 percent and *TR decreases.*
- If demand is unit elastic, a 1 percent price cut increases the quantity sold by 1 percent and so *TR does not change.*

Calculating the change in total revenue

Find the change in total revenue using the following two points along a demand curve: ($20, 150) and ($10,350).

$$TR_A = \$20 \text{ x } 150 = \$3{,}000.$$

$$TR_B = \$10 \text{ x } 350 = \$3{,}500.$$

$$\Delta TR = TR_B - TR_A = \$3{,}500 - \$3{,}000 = \$500$$

- Using our previous answer, these points correspond with an elastic demand curve.
  - It is expected that the change in total revenue will be positive since price is falling.

Expenditure and Elasticity

- If demand is elastic, a 1 percent price cut increases the quantity sold by more than 1 percent and *expenditure on the item increases*
- If demand is inelastic, a 1 percent price cut increases the quantity sold by less than 1 percent and *expenditure on the item decreases*
- If demand is unit elastic, a 1 percent price cut increases the quantity sold by 1 percent and so *expenditure on the item does not change*

Determinants of price elasticity of demand

- Consumers are more sensitive to price changes for some goods and services than for others.
- Many factors determine consumers’ responsiveness to price changes.
  - Closeness of substitutes
  - Proportion of income spent on good
  - Time elapsed since price change

Closeness of substitutes

- *The closer the substitutes for a good or service, the more elastic is the demand for it.*
- *If a close substitute exists, then even a small rise in the price will see a large fall in the quantity demanded, vice versa.*

- Narrowness: The elasticity of demand for PCs is low, but the elasticity of demand for Dell or HP is high.
- Necessities such as food and insulin have poor alternatives and thus have an inelastic demand. Luxurious goods such as cars and vacations have an elastic demand.

Proportion of income spent on the good

The greater the proportion of income spent on a good, the more elastic is the demand for it.

- Your demand for housing is more elastic than your demand for toothpaste. (think of how much each of them takes away from your income)
- For goods that use up a large proportion of our income and budget, a small price change can make them unaffordable e.g. housing rent

Time elapsed since price change

The longer the time has elapsed since a price change, the more elastic is the demand for it.

The more time consumers have to adjust to a price change, the more elastic is the demand for that good.

Cross-price elasticity of demand

- The *cross-price elasticity of demand* is a measure of how the quantity demanded of one good changes when the price of a different good changes.
- The midpoint formula calculates the elasticity between the quantity demanded of good A and the price of good B:

$$\varepsilon_{A,B} = \frac{\%\Delta Q_A}{\%\Delta P_B} = \frac{(Q_2 - Q_1)/[(Q_2 + Q_1)/2]}{(P_2 - P_1)/[(P_2 + P_1)/2]}$$

- The cross-price elasticity of demand can be positive or negative.
  - $\varepsilon_{A,B} > 0$: the two goods are substitutes.
  - $\varepsilon_{A,B} < 0$: the two goods are complements.

Cross-price elasticity of demand

The magnitude of cross elasticity of demand depends on the strength of the relationship between the two goods.

- In case of substitutes:
- If XED > 1: high elasticity because the two goods are very close substitutes e.g. coke and pepsi
- If XED < 1: low elasticity the two goods are weak substitutes e.g. burgers and pizzas
- In case of complements:
- If XED < - 1: high elasticity because the two goods are very close complements e.g. pencil and sharpener
- If XED > - 1: low elasticity because the two goods are weak complements e.g. tea and milk
- If goods are unrelated: XED = 0 e.g. TVs and shirts

Income elasticity of demand

- The *income elasticity of demand* is a measure of how much the quantity demanded changes in response to a change in consumers' incomes.
- The midpoint formula calculates the elasticity between the quantity demanded of a good and consumer's income:

$$\varepsilon_I = \frac{\%\Delta Q}{\%\Delta I} = \frac{(Q_2 - Q_1)/[(Q_2 + Q_1)/2]}{(I_2 - I_1)/[(I_2 + I_1)/2]}$$

- The income elasticity of demand can be positive or negative.
  - $\varepsilon_I > 0$: the good is normal. If $\varepsilon_I > 1$, then it is a luxury.
  - $\varepsilon_I < 0$: the good is inferior.

Income elasticity of demand

If normal goods have

- Income elasticity of demand > 1: As income increases, the quantity demanded increases faster than income i.e. luxury goods such as international travel, jewelry

- Income elasticity of demand < 1: As income increases, the quantity demanded increases slower than income i.e. necessity goods such as newspapers, clothing

- As you get richer, you spend a larger proportion of your income on luxury goods and a smaller proportion of your income on necessary goods.

Calculating the income elasticity of demand

- When consumers’ income is $100, they buy 250 units of a good. When consumers’ income increases to $200, they buy 500. Find the income elasticity of demand using the *midpoint formula*.
- Is the good inferior or normal?
- Is it a luxury good?

Price elasticity of supply

- The concept of elasticity can also be applied to supply.
- The *price elasticity of supply* measures producers' response (in quantity) to a change in price.
  - Uses same midpoint formula but replaces quantity demanded with quantity supplied.
  - Elasticity is always positive.
  - Same interpretation:
    - Elastic: $\varepsilon_s > 1$.
    - Unit Elastic: $\varepsilon_s = 1$.
    - Inelastic: $\varepsilon_s < 1$.

Determinants of price elasticity of supply

Many factors determine producers' responsiveness to price changes.

1. Resource substitution possibilities

If a good can only be produced by using rare or unique resources or resources that are not easily available, it will have a low elasticity of supply i.e. the good will have a low responsiveness to price changes.

2. Time frame for supply decisions

Goods that take a long time to produce have a very low price elasticity of supply. On the other hand, goods that can be quickly and easily made have high elasticity of supply. In the long run, supply is more elastic.

Examples

1. Good weather brings a bumper tomato crop. The price falls from £7 to £5 a load, and the quantity demanded increases from 300 to 500 loads a day. Over this price range,

a. What is the price elasticity of demand?

b. Describe the demand for tomatoes.

Examples

2. A fall in the price of X from $12 to $8 causes an increase in the quantity demanded of X from 500 to 700 units and quantity demanded of Y from 900 to 1,100 units.

a. What is the cross elasticity of demand between X and Y?

b. What is the relationship between goods X and Y?

Examples

3. The demand function for an energy drink on an exam day is given by $P = 130 - 2Qd$

a. Assume you sell one litre of the drink at Tk 50. What is your total revenue earned?

b. Hoping to make more profits, you decide to increase the price to TK 60 per litre. How does your total revenue change?

c. Over this price range, what is the PED for the energy drink?

Summary

- *Key concepts:*
- Price elasticity of demand
- Elastic vs Inelastic
- Cross-price elasticity of demand
- Income elasticity of demand
- Price elasticity of supply

- *Reading: Chapter 4 of "Economics" by Parkin, Powell, Mathews 6th Ed.*